LiViFuser-Nav

Geometry-Aware Dense Vision + Sparse 2D LiDAR Fusion
for Uncertainty-Aware Local Navigation on Low-Cost Robots

Architecture Specification v1.1
Final Locked for Pilot Implementation

STATUS: FINAL LOCKED — READY FOR PILOT IMPLEMENTATION

Platform	TurtleBot3 Burger (differential drive)
Sensors	LDS-03 (2D LiDAR 360°) + Raspberry Pi Camera Module v3
Vision Backbone	Frozen DINOv3-S+/16
Policy	Heteroscedastic ACT (H=8) + spatial cross-attention decoder
Uncertainty	Aleatoric (σ) + Epistemic/OOD (Mahalanobis on DINO features)
Control	10 Hz target | Receding-horizon (1-step) | Independent safety supervisor
Date Locked	2026-07-27

This is the final locked specification. It incorporates geometry-aware fusion with spatial structure preserved to the action head, heteroscedastic uncertainty with variance-collapse protection, free Mahalanobis OOD gating on frozen DINOv3 features, mandatory goal conditioning, receding-horizon execution, and an independent safety supervisor.

1. Research Positioning & Contribution

1.1 Contribution Statement

A geometry-aware gated fusion policy that combines dense self-supervised visual tokens with full-surround sparse 2D LiDAR, dual uncertainty signals (aleatoric + free Mahalanobis OOD), goal-conditioned temporal memory, and safety-constrained receding-horizon execution for real-time navigation on low-cost differential-drive robots.

1.2 Differentiation

Work	Focus	Our Differentiation
ViLiNT (2026)	RGB + 3D LiDAR, diffusion, larger robots	Sparse 2D LiDAR + dense SSL vision on small robots; FOV-masked geometry-aware fusion; dual uncertainty; explicit safety
NoMaD / ViNT	Vision-only foundation models	Structured sparse LiDAR fusion + uncertainty + goal-conditioned temporal memory
Classical Nav2	Hand-engineered local planners	Fully learned local policy with modern dense features and calibrated uncertainty

1.3 Explicitly Not Claimed

- Gated cross-attention itself is not claimed as novel.
- Aleatoric σ alone is not claimed to detect OOD; Mahalanobis supplies the epistemic signal.
- Contribution is the complete system formulation + empirical validation on low-cost hardware, including the finding that frozen SSL features can serve as a free real-time OOD gate.

2. Final Architecture (v1.1)

2.1 End-to-End Data Flow

```
RGB image
■
▼
Frozen DINOv3-S+/16
■■■ Spatially pooled visual tokens (32–64) + pooled vector z for Mahalanobis

LiDAR scan (LDS-03)
■
▼
1D CNN + range/validity + angular positional encoding
■■■ LiDAR tokens (72–90)
■■■ FOV / front beams → Geometry-aware Cross-Attention with visual tokens
■■■ Out-of-FOV beams → Separate LiDAR-only residual branch

Goal [ρ, sin(α), cos(α)] → Goal MLP → Goal embedding
Robot state (v, ω, …) → State embedding

Current-step fused token set T_fused
■
■■■■ Pool(T_fused) + goal + state → GRU Temporal Memory (K=8) → h_t
■
■■■■ ACT Decoder cross-attends over T_fused
with h_t as additional context token
■
▼
Heteroscedastic ACT Head (H=8)
y ∈ R^{H×4} = [μ_v, μ_ω, log σ²_v, log σ²_ω]
■
▼
Receding-Horizon Controller (execute 1 step, replan)
■
▼
Independent ROS 2 Safety Supervisor
Inputs: policy command, LiDAR, σ (aleatoric), D_Mahalanobis (OOD)
```

2.2 Component Specifications

Component	Choice	Key Details
Vision Encoder	Frozen DINOv3-S+/16	No fine-tuning in v1. Spatial structure preserved.
Visual Tokens	32–64 after fixed pooling	Fixed spatial pooling first (easier to debug).
LiDAR Encoder	1D CNN + angular PE	Input [r, sin θ, cos θ, validity]. 72–90 tokens.
Fusion	Geometry-aware Cross-Attn + Gate	FOV mask from extrinsics. Rear = LiDAR-only path.

Component	Choice	Key Details
Temporal path	Pooled features → GRU (K=8)	Memory of recent fused observations + goal + state.
Action decoder	Cross-attend over T_fused + h_t	Spatial structure reaches the policy head.
Policy Head	Heteroscedastic ACT, H=8	Mean + log-variance. No CVAE in v1.
Goal	Mandatory relative waypoint	g = [ρ, sin α, cos α] in robot frame.
Execution	Receding horizon, 1-step	2-step only after latency is proven.
Safety	Independent ROS 2 node	Higher authority. Uses σ and Mahalanobis D.

2.3 Initial Dimensions

Component	Value
DINOv3 backbone	S+/16, frozen
Visual tokens after reduction	32–64
LiDAR tokens	72–90
Common feature dimension	256
Cross-attention heads	4
Goal / state embedding	32 each
GRU hidden dimension	256
Temporal context K	8
Action horizon H	8
Executed steps per replan	1 (initially)
Output shape	H × 4

3. Geometry-Aware Fusion & Spatial Path to Actions

3.1 LiDAR Token Representation

$$\text{For each LiDAR sector/beam } i\text{: } l_i = [\ r_i,\ \sin\theta_i,\ \cos\theta_i,\ m_i\ ]$$

r = normalized range, θ = angle, m = validity mask. Group nearby beams into 72–90 angular sectors.

3.2 Camera–LiDAR Geometric Mask

Using calibrated intrinsics and extrinsics, classify each LiDAR token as inside or outside the camera FOV. Front/FOV beams attend to compatible visual patches via masked cross-attention. Out-of-FOV beams use a separate LiDAR-only feed-forward path — preventing rear obstacles from being incorrectly associated with front-camera features.

3.3 Gated Fusion

$$z_{VL} = \mathrm{CrossAttn}(T_L^{front},\ T_V,\ T_V)$$
$$g = \sigma(\ \mathrm{MLP}_g(\ \mathrm{Pool}(z_{VL})\ \blacksquare\ \mathrm{Pool}(T_L)\ )\ )$$
$$f^{fusion} = g\ \blacksquare\ \mathrm{Pool}(z_{VL}) + (1-g)\ \blacksquare\ \mathrm{Pool}(T_L) + \mathrm{Pool}(T\blacksquare_L^{rear})$$

The current-step fused token set T_fused is retained (not discarded after pooling). Pool(T_fused) feeds the GRU for temporal memory. The ACT decoder cross-attends over T_fused with the GRU hidden state h_t as an additional context token. This ensures geometry-aware spatial structure actually reaches the action outputs.

3.4 Goal & State Integration

$$e_{g,t} = \mathrm{MLP}_g(\rho_t,\ \sin\alpha_t,\ \cos\alpha_t)$$
$$x_t = [\ \mathrm{Pool}(T\_fused);\ e_{g,t};\ e_{s,t}\ ]$$
$$h_t = \mathrm{GRU}(x_t,\ h_{t-1})$$

Goal embedding is introduced before the GRU so temporal memory conditions on navigation intent.

4. Dual Uncertainty & Training Loss

4.1 Output Parameterization

$$y_{t,h} = [\ \mu_{v,h},\ \mu_{\omega,h},\ \log\sigma^2_{v,h},\ \log\sigma^2_{\omega,h}\ ]$$

Predicted σ captures **aleatoric** uncertainty (ambiguity/noise among training-like situations). No CVAE is used in v1 — a deterministic chunking head + heteroscedastic NLL is cleaner.

4.2 Mahalanobis OOD (Epistemic) Signal

After data collection, extract pooled DINOv3 feature $z \in R^d$ for every training frame. Fit a Gaussian (μ, Σ) with shrinkage or diagonal covariance for stability. At deployment compute:

$$D(z) = \sqrt{(\ (z-\mu)^T\ \Sigma^{-1}\ (z-\mu)\ )}$$

Large D means the current visual features lie far from the training distribution (OOD / epistemic signal). Cost is negligible (one matrix-vector product). Both σ and D are fed to the safety supervisor. This converts frozen SSL features into a free real-time OOD detector for closed-loop safety gating.

4.3 Training Loss with Variance-Collapse Protection

Warmup phase (first N epochs): pure L1 or MSE on action means only.
Main phase: heteroscedastic Gaussian NLL

$$L_{NLL} = \tfrac{1}{2}\ \Sigma_h\ \Sigma_{d\in\{v,\omega\}}\ [\ \exp(-s^d)\ (a^d - \mu^d)^2 + s^d\ ]$$

where $s = \log\sigma^2$. Clip $s \in [-5, 2]$ early for numerical stability. Warmup prevents the known failure mode of variance collapse on hard samples. Optional: temporal smoothness penalty on consecutive μ in the chunk; mild gate regularization.

Full v1 objective: $L = \lambda_{NLL}\ L_{NLL} + \lambda_{smooth}\ L_{smooth} + \lambda_{gate}\ L_{gate}$
No CVAE / no KL term in v1.

5. Execution & Safety Supervisor

5.1 Receding-Horizon Execution

Predict H=8 future actions; execute only the first command; obtain new observation; replan. One-step execution is the default for fastest reaction to people, new obstacles, and sensor dropouts. Two-step execution only after latency measurements prove the system can sustain the desired rate.

5.2 Safety Supervisor Hierarchy (Higher Authority than Policy)

$$a_{executed} = \mathrm{SafetyFilter}(\ a_{policy},\ z_{LiDAR},\ \sigma,\ D_{Mahalanobis},\ \Delta t_{sensor}\ )$$

1. Emergency stop or manual override

2. Sensor or command timeout stop
3. Collision-distance / braking-distance safety stop
4. Velocity and acceleration limiting
5. Uncertainty-based slowdown or stop (uses both σ and D)
6. Learned-policy command (lowest priority)

5.3 Dual-Signal Uncertainty Rule

u combines aleatoric σ and epistemic D. Example form:

$$v_t^{safe} = \gamma(\sigma, D) \cdot v_t^{policy} \text{ with } 0 \le \gamma \le 1$$

Thresholds chosen via validation risk–coverage analysis on held-out and novel environments. Key paper figure: collision / intervention rate vs. D-threshold, showing the OOD gate catches failures that aleatoric σ alone misses.

6. Latency Constraint (Critical)

Target control rate: **10 Hz**. Full pipeline (capture → preprocess → inference → safety → publish) must stay under **100 ms** (ideal ≤ 80 ms average). DINOv3-S+/16 runs on the arranged GPU (not on the Raspberry Pi). Measure full round-trip latency early, including any network hop. If marginal, further reduce visual tokens or stay strictly at one-step execution.

7. Data Collection Plan

7.1 Recording Schema (@ 10 Hz)

Field	Source	Notes
rgb	RPi Cam v3	Train resolution 224×224 (or 320×240 capture)
lidar	LDS-03	Ranges + angles; ~5 Hz spin → nearest-scan association
action	Teleop	linear.x, angular.z
robot_state	Odom	Current v, ω
goal	Planner / operator	Relative [ρ, sin α, cos α] — mandatory
timestamp	System	Required for multi-rate association

7.2 Protocol Highlights

- Primary store: rosbag2. LeRobot format as exported training view.
- LDS ~5 Hz vs 10 Hz sampling → nearest-timestamp association (expect duplicated scans).
- Report total time, distance, turns, obstacle density, lighting, interventions, near-collisions.
- Train/val/test split by physical environment (not random frames) to avoid leakage.
- 150–200 episodes for first working policy; 300–500 for stronger results.
- Short Gazebo/Isaac sanity check recommended before large-scale real teleoperation.
- Drive smoothly (joystick preferred); deliberately include near-misses and recoveries.

8. Baselines & Evaluation

8.1 Required Baselines

- Classical Nav2 local navigation
- LiDAR-only learned policy
- RGB-only DINO policy
- Simple RGB–LiDAR concatenation (no cross-attention)
- Cross-attention without gating / without FOV mask
- Full model without temporal memory
- Full model without uncertainty (σ or Mahalanobis)
- Aleatoric-only vs aleatoric + Mahalanobis safety gating

8.2 Metrics

Navigation: Success Rate, Collision Rate, path efficiency, recovery success.
Uncertainty: Failure-prediction AUROC, NLL, calibration error, risk–coverage curves, collision rate at different σ and D thresholds.
Systems: End-to-end latency, achieved control frequency, GPU/RAM, parameter count.

9. Pilot Implementation Roadmap

Stage	Focus	Deliverable
1	Infrastructure	ROS 2 recording, camera–LiDAR calibration, goal-conditioned export, latency measurement
2	Core model	Frozen DINO features, LiDAR encoder, geometry-aware fusion, ACT + spatial decoder
3	Reliability	GRU memory, heteroscedastic head (with warmup), Mahalanobis fit, receding-horizon controller
4	Safety & deploy	Independent safety supervisor (σ + D), closed-loop real-robot tests
5	Research polish	Baselines, ablations, latency report, risk–coverage and D-threshold figures

10. Locked Decisions Summary (v1.1 Final)

Decision	Final Choice
Vision backbone	Frozen DINOv3-S+/16 only
Visual tokens	32–64 after fixed spatial pooling
LiDAR tokens	72–90 with angular + validity encoding
Fusion	Geometry-aware Cross-Attn (FOV-masked) + gated residual + LiDAR-only rear
Spatial path to actions	ACT decoder cross-attends over current T_{fused}; GRU h_t as context
Goal conditioning	Mandatory relative waypoint [ρ, sin α, cos α]
Temporal memory	GRU, K = 8 (pooled path)
Policy	Heteroscedastic ACT, H = 8; no CVAE / no KL in v1
Loss	L1/MSE warmup → full heteroscedastic NLL (+ optional smooth/gate)
Aleatoric uncertainty	Predicted σ from heteroscedastic head
Epistemic / OOD	Mahalanobis distance on pooled frozen DINO features
Execution	Receding horizon, 1-step first
Safety	Independent ROS 2 supervisor; priority hierarchy; uses σ and D
Data	rosbag2 primary + LeRobot export; environment-based splits; nearest-scan association
Control rate	10 Hz target; full pipeline < 100 ms
Status	v1.1 — Final Locked for Pilot Implementation

Immediate next actions: (1) synchronized ROS 2 recording + camera–LiDAR calibration, (2) latency measurement on the arranged GPU, (3) LiDAR-only and concatenation baselines, (4) geometry-aware fusion + spatial ACT training with warmup, (5) Mahalanobis fit on training features, (6) closed-loop deployment behind the safety supervisor.

LiViFuser-Nav Architecture Specification v1.1 — Final Locked Document
Single source of truth for pilot implementation and research documentation.